

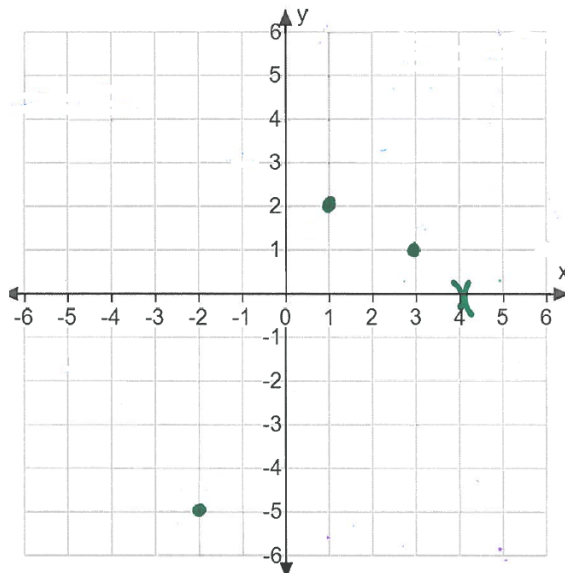
Problem 12

Problem 12

Problem 1 Sketch a possible graph of a function that satisfies all of the given properties. (You do not need to find a formula for the function.)

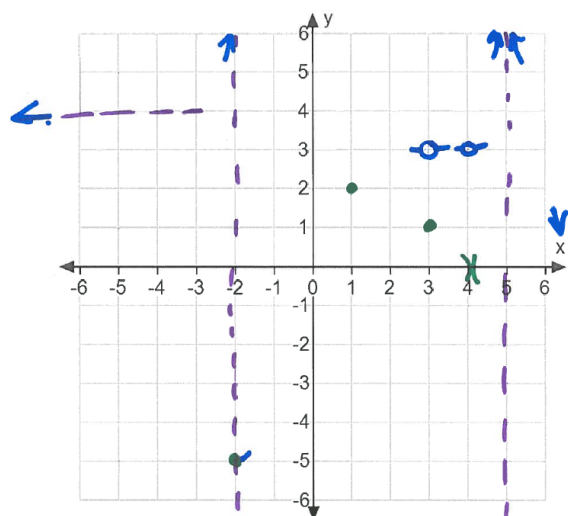
$$\begin{array}{lll} \lim_{x \rightarrow -2^-} f(x) = \infty & f(-2) = -5 & f(1) = 2 \\ \lim_{x \rightarrow \infty} f(x) = -\infty & \lim_{x \rightarrow -\infty} f(x) = 4 & \lim_{x \rightarrow 5} f(x) = \infty \\ \lim_{x \rightarrow 3} f(x) = 3 & f(3) = 1 & \\ \lim_{x \rightarrow -2^+} f(x) = -5 & f(4) \text{ is undefined} & \lim_{x \rightarrow 4} f(x) = 3 \end{array}$$

Explanation. There are many correct solutions to this problem. One possible solution can be constructed as follows: First we draw the given points. We'll also draw open brackets on the x-axis at $x = 4$ since $f(x)$ is not defined at $x = 4$ (This is seen in green.)



Then we draw the vertical and horizontal asymptotes (in purple) and arrows indicating how the graph approaches the asymptote (in blue). We also draw end-behavior as x approaches ∞ and $-\infty$ (in blue). We can also draw tails where we know limits at particular x-values. Such as $\lim_{x \rightarrow -2^+} f(x) = -5$.

Problem 12



Finally we connect our graph together (seen in blue)

